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Physlib

A community-run library of formalized physics

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github.com/leanprover-community/physlib josephtoobysmith.com

Overview

1. The formalization landscape
2. Physlib: the details
3. How to contribute?

The formalization landscape

The formalization landscape
Physlib: the details
How to contribute?

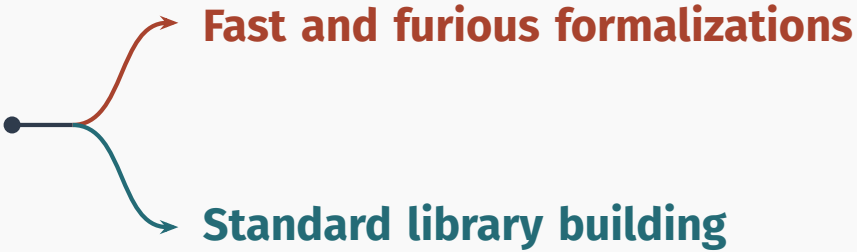
A shift in activities

Back in 2023, within Lean

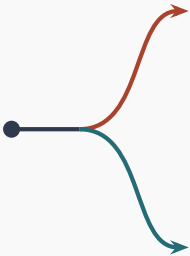
Formalization was synonymous with building a **community run standard library** (corpus of knowledge), **Mathlib**…

(By standard library, or corpus of knowledge, I mean a high quality collection of results.)

Since then…


Fast and furious formalizations
Standard library building

These are mutually reinforcing, with different purposes



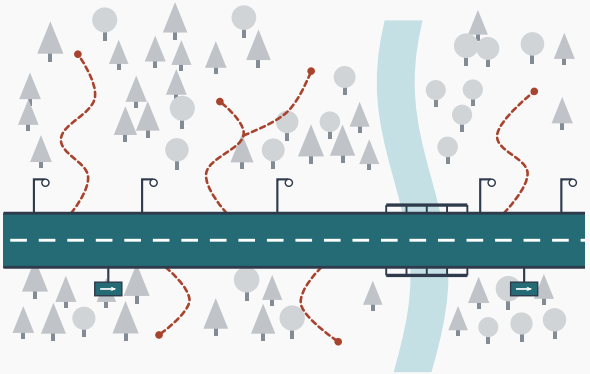
Fast and furious formalizations

Show correctness of existing or new results, while aiming to discover new things by **quickly pushing forward the frontier**.

Standard library building

Create a readable, **human trusted**, **high quality** corpus of knowledge containing the best results possible in a connected and reusable way. In doing so, forming a **solid foundation** for future formalizations to build on, and leading to discovery by comprehensively covering areas with high quality results, allowing us to see new connections, reductions and methods.

An analogy



Fast and furious formalization

Like **little paths** through the forest: **quick to make**, but sometimes not trusted and hard to follow.

Standard library building

Like building a **road**, with lots of **infrastructure**.

The future

For formalization in physics to succeed we still need **both** of these activities.

Without standard libraries, fast and furious formalizations **lose trust**, **have to remake results**, **become incompatible**, etc.

The project **Physlib** serves as the **standard library for physics**.

The three main standard libraries

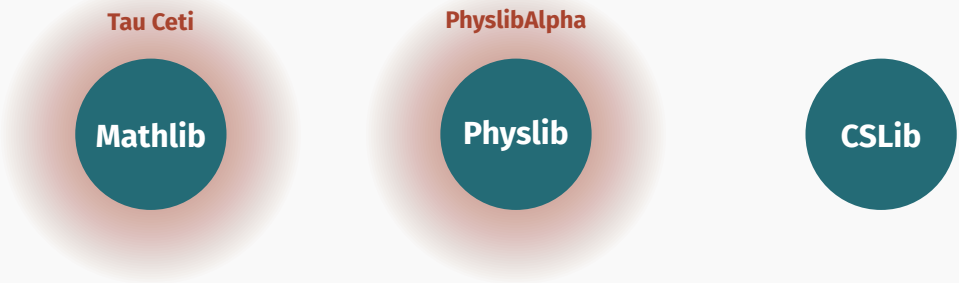


Helping connect to these libraries…

Fast and furious halo projects



The three main standard libraries



Physlib: the details

The formalization landscape

Physlib: the details

How to contribute?

Physlib



leanprover-community / Physlib

A community-driven corpus of knowledge of results from physics, **formalised into Lean 4** where possible.

public

lean 4

78
open PRs (∼5 new a day)

120
contributors

340
people on the Zulip

818 stars **218** forks started 2024

The goal of Physlib

To be a **high quality**, **human readable**, **trusted**, **community run** standard library of results from physics in Lean.

Trust from human review

Like with **Mathlib** and **CSLib**, everything going into **Physlib** is **heavily peer-reviewed**.

Giving trust on **physics correctness**, **quality** and **readability**, not just mathematical correctness.

Areas covered

Top-level directories of `Physlib/`, by lines of Lean.

Relativity	27,545	StatisticalMechanics	2,449
QFT	20,162	ProbabilisticTheory	2,306
Particles	19,104	Cosmology	1,858
SpaceAndTime	16,069	Meta	1,773
Mathematics	15,795	FluidDynamics	1,361
QuantumMechanics	13,057	CondensedMatter	1,003
ClassicalMechanics	11,784	ClassicalFieldTheory	554
Electromagnetism	9,442	Thermodynamics	432
StringTheory	5,285	Optics	48
Units	4,076	LatticeQFT	22

154,125 lines of Lean across **619** files.

Some example results

A selection of results in Physlib, with the directory each sits in.

Wick's theorem	QFT
Results from SU(5) F-theory	StringTheory
Standard Model Lagrangian	Particles
Maxwell's equations	Electromagnetism
Two Higgs doublet model	Particles
Grand unified theories	Particles
Anomaly cancellation	Particles

Quantum harmonic oscillator	QuantumMechanics
Set up for QED and general gauge theory	PhyslibAlpha
Tight binding model	CondensedMatter
Ideal gas law	StatisticalMechanics
Lagrangian systems	ClassicalMechanics
Probabilistic theories	ProbabilisticTheory

Over **7,900** theorems and lemmas in total.

Parts of the philosophy of the project

1. **Community run**, and **community led**.
2. Best corpus is a **monolithic** one, whilst keeping **independence over areas** (anyone can review).
3. Allow **any starting point**, but don't exclude axiomatizations.
4. Don't be scared of **refactoring**.

Physlib's halo

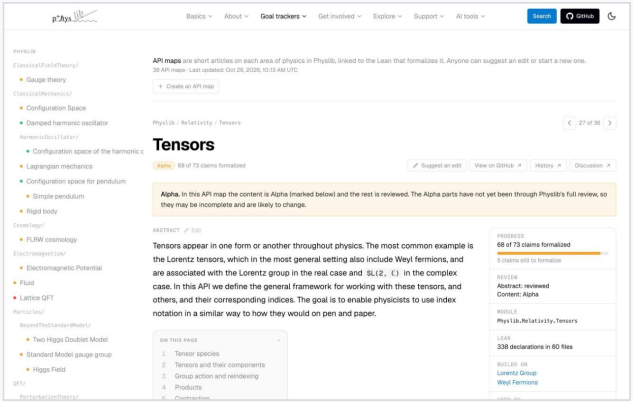


Allows for **fast reusable formalization** in physics.

With a **"one-look"** review policy.

The Physlib Atlas (API-maps)

Human-readable pages which allow an **entry point** for physicists, and a **bridge** through which they can contribute…



section 3 of 3

How to contribute?

The formalization landscape
Physlib: the details
How to contribute?

Activities one could do

Review

e.g. reviewing PRs in your area.

Maintenance

e.g. adding new skills, linters, etc.

Refactoring

e.g. refactoring tensors for index notation.

Foundational work

e.g. foundations of string theory.

Formalization

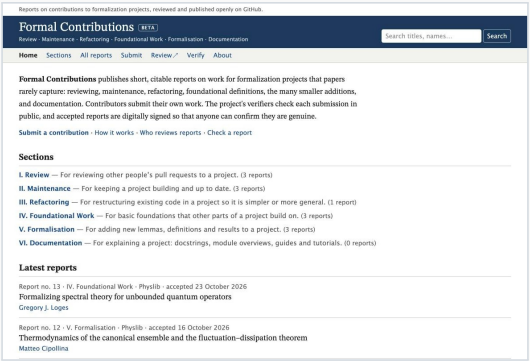
e.g. extending the formalization of invariance of the SM.

Documentation

e.g. helping add an Atlas in your area of expertise.

Credit for contributions

formalcontributions.com



Physics 2.0

A plan for the future of physics

Contributions to projects like **Physlib**, with credit given from sites like **formalcontributions.com**, could make a **new landscape** for how we do physics.

Thank you

Happy to discuss how we can make this project better serve the needs of the community

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github.com/leanprover-community/physlib formalcontributions.com josephtoobysmith.com